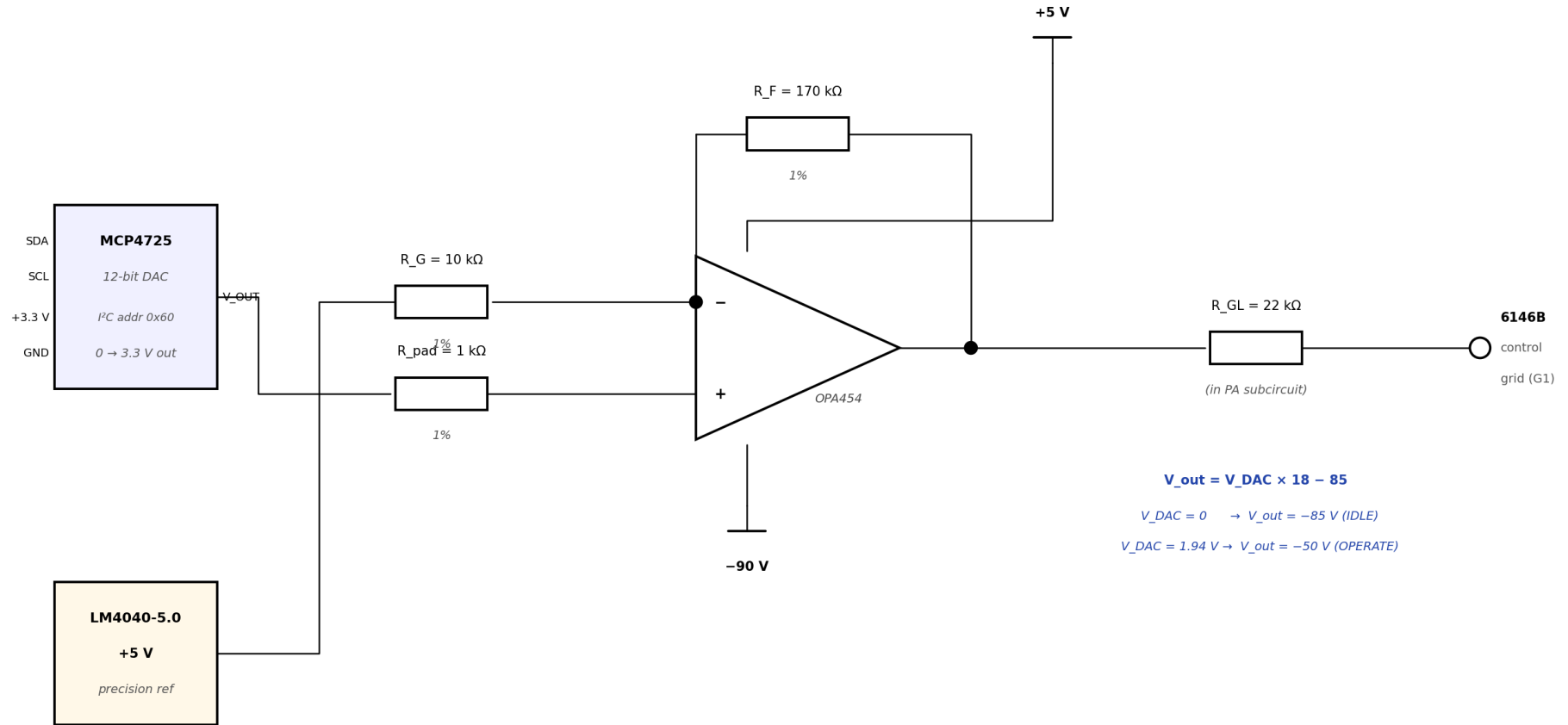


Design A — Per-Tube 6146B Grid Bias with OPA454 HV Op-Amp

$V_+ = +5\text{ V}$, $V_- = -90\text{ V}$ (95 V total, within OPA454 100 V operating max). $V_{\text{out}} = -85\text{ V}$ (IDLE) $\rightarrow -50\text{ V}$ (OPERATE).



-90 V rail: small isolated DC-DC + voltage doubler (e.g., Murata NMA0509 $\rightarrow \pm 9\text{ V} \rightarrow 10\times$ doubler stages),
regulated to -90 V via a TL431 shunt or 91 V zener. ~5 mA total for both PA channels.
+5 V rail: same +5 V supply as the LM4040 reference and DAC logic; 100 nF + 10 μ F decoupling at each OPA454 supply pin.
Replicate this channel for the second tube (separate I²C address; shared rails OK).

Figure 1. Design A — OPA454 HV op-amp version. One channel; replicate for the second tube.

Design A — OPA454 Components & Transfer Function

Ref	Value / Part	Notes
DAC	MCP4725 (×2)	I ² C 12-bit; EEPROM startup = 0 → output = 0 V → V _{out} = −85 V (IDLE)
Op-amp	OPA454 (×2)	HV op-amp, V ₊ = +5 V / V _− = −90 V (95 V total, within 100 V op max)
V _{REF}	LM4040DIZ-5.0	Precision +5 V shunt reference
R _{pad}	1 kΩ 1%	DAC → (+) input series; impedance matching
R _G	10 kΩ 1%	(−) input ↔ V _{REF} (+5 V)
R _F	170 kΩ 1% (or 169 kΩ E96)	Feedback (−) ↔ V _{out}
Decoupling	100 nF X7R + 10 μF on each rail at the OPA454	
+5 V rail	Same as DAC / LM4040 supply	Trivial current; share existing rail
−90 V rail	Isolated DC-DC (e.g., Murata NMA0509) + voltage doubler PA1431-1	5-stage double-PA1431-1 shunt regulator
R _{GL}	22 kΩ (already in PA subcircuit)	Unchanged

Transfer Function

$$V_{\text{out}} = V_{\text{DAC}} \times (1 + R_{\text{F}}/R_{\text{G}}) - V_{\text{REF}} \times R_{\text{F}}/R_{\text{G}} = V_{\text{DAC}} \times 18 - 85$$

$V_{\text{DAC}} = 0 \text{ V} \rightarrow V_{\text{out}} = -85 \text{ V}$ (IDLE: tubes deeply cut off, zero plate current)

$V_{\text{DAC}} = 1.94 \text{ V} \rightarrow V_{\text{out}} = -50 \text{ V}$ (OPERATE: shallow class C, ready for envelope) — DAC code 2410 (of 4095)

$V_{\text{DAC}} = 3.3 \text{ V} \rightarrow V_{\text{out}} = -25.6 \text{ V}$ (*firmware MUST cap below this* — would forward-bias the grid)

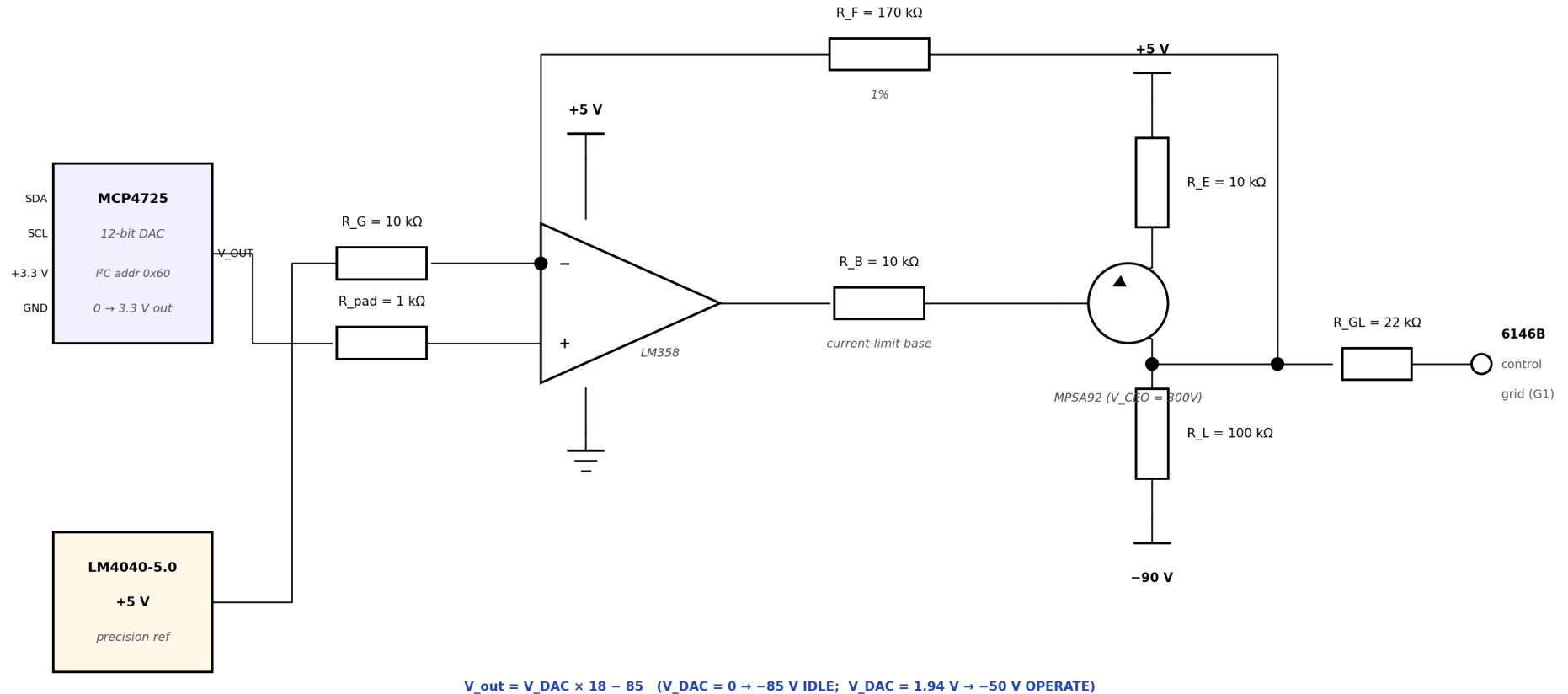
Common-mode input range check: with V₊ = +5 V / V_− = −90 V and OPA454's typical 3 V CM margin, the (+) input range is roughly −87 V to +2 V. V_{DAC} at the (+) input swings 0 to ~2 V (firmware-bounded), inside the range with margin.

Output swing check: rail-to-rail−2 V gives output range −88 V to +3 V. Comfortably covers −85 V (IDLE) and −50 V (OPERATE).

Supply margin: the +5 V / −90 V choice gives 95 V total — 5 V under the OPA454 100 V operating max, well under the 105 V absolute max.

Design B — Per-Tube 6146B Grid Bias with Discrete HV PNP Level-Shifter

LM358 low-voltage op-amp + MPSA92 PNP ($V_{CEO} = 300\text{ V}$). Same transfer function as Design A. No op-amp voltage limit.



PNP collector V_{CE} ranges: $V_E = +4.4\text{ V}$ (V_{BE} drop), V_C ranges -85 to $-50\text{ V} \rightarrow V_{CE_max} \approx 89\text{ V}$, well within MPSA92's $300\text{ V } V_{CEO}$ rating.

Max collector current: $(V_{out} - V_L) / R_L = (-50 - (-90)) / 100\text{ k} = 0.4\text{ mA}$. Max $P_{diss} = V_{CE} \times I_C \approx 35\text{ mW}$. Anything in TO-92 works.

Op-amp output swing only needs $\pm 0.7\text{ V}$ to drive the PNP base around its operating point. LM358, TL072, OP07 all work; LM358 chosen for cost and single-supply.

Same -90 V rail and same $+5\text{ V}$ rail as Design A. Replicate channel-per-tube.

Figure 2. Design B — LM358 + discrete MPSA92 PNP level-shifter. Same transfer function as Design A; no op-amp voltage limit.

Design B — Discrete PNP Components & Notes

Ref	Value / Part	Notes
DAC	MCP4725 (×2)	Same as Design A
Op-amp	LM358 (×2; one half each), or TL072	LV op-amp, single +5 V supply — no HV constraint
HV transistor	MPSA92 (×2)	PNP, $V_{CEO} = 300\text{ V}$, TO-92; alternative: KSP44, ZTX560, 2N5401 (160 V)
V_REF	LM4040DIZ-5.0	Same +5 V shunt reference as Design A
R_pad	1 kΩ 1%	DAC → (+) input series
R_G	10 kΩ 1%	(-) input ↔ V_REF (+5 V)
R_F	170 kΩ 1% (or 169 kΩ E96)	Feedback from PNP collector ↔ (-) input
R_B	10 kΩ	Base current limit; sets loop slew comfortably
R_E	10 kΩ	Emitter resistor to +5 V; sets ~0.4 mA emitter current
R_L	100 kΩ	Collector pull-down to -90 V; sets load impedance
Decoupling	100 nF X7R at each LM358 supply pin; 1 nF NP0 across each transistor B-E for HF stability	
+5 V rail	Shared with DAC, LM4040, LM358	
-90 V rail	Same isolated DC-DC as Design A	Loads R_L (×2 tubes); ~1 mA total
R_GL	22 kΩ (already in PA subcircuit)	Unchanged

Transfer Function and Operation

Same as Design A: $V_{out} = V_{DAC} \times 18 - 85$

The PNP collector node is the output. LM358 closes the loop via R_F sensing the collector voltage; the op-amp drives the PNP base to whatever level forces $V_{(-)} = V_{(+)} = V_{DAC}$. No supply-voltage limits on the loop because the HV stress is borne entirely by the PNP transistor.

PNP stress check: $V_{CE_max} \approx V_E - V_{C_min} = +4.4\text{ V} - (-85\text{ V}) = 89\text{ V}$ — well below MPSA92's 300 V V_{CEO} rating (3.4× margin). $\text{Max } I_C = (V_{out} - V_{neg}) / R_L = (-50 - (-90)) / 100\text{ k}\Omega = 0.4\text{ mA}$. $\text{Max } P_{diss} = V_{CE} \times I_C \approx 35\text{ mW}$ (any TO-92 fits comfortably).

Op-amp swing required: the LM358 only needs to swing about $\pm 0.7\text{ V}$ around the PNP base operating point — fully within single-supply LM358 specs. Even at the IDLE end ($V_{out} = -85\text{ V} \rightarrow I_C \approx 0$), the op-amp output stays within rails.

Why this exists: if the OPA454 is unobtainable, of dubious authenticity (the marketplace has counterfeits), or you prefer not to operate any single part at its supply max, the discrete design is electrically equivalent and built from in-stock generics. The MPSA92 is a \$0.10 part in singles.

Shared: Two-State Operation, Power-Up, Fail-Safe, Layout

Two-State Operation

Bias has two firmware-managed states with smooth transitions:

IDLE (key-up): $DAC = 0 \rightarrow V_{out} = -85\text{ V} \rightarrow$ 6146B fully cut off, plate current = 0, standby dissipation = 0 W per tube. This is the safe-park state.

OPERATE (key-down): $DAC = 2410$ (of 4095) $\rightarrow V_{out} = -50\text{ V} \rightarrow$ shallow class C, smooth conduction onset, full envelope dynamic range available to the predistortion LUT.

Per-tube trim: the OPERATE DAC code may differ by a few counts between tubes to balance for tube-to-tube spread. Calibrated at bring-up via the cathode-current monitor.

Power-Up Sequence and Firmware Timing

Power-on: Rails come up (+5 V, -90 V). MCP4725 EEPROM startup value = 0 $\rightarrow DAC = 0\text{ V} \rightarrow V_{out} = -85\text{ V}$ on both tubes. PA is in deep cutoff before the MCU finishes booting.

Key-down sequence: (1) WinKey signals key-down; (2) per-tube bias DACs commanded IDLE $\rightarrow$ OPERATE (each $\sim 25\text{ }\mu\text{s}$ over I²C); (3) wait 200 μs for op-amp + grid input cap to settle; (4) launch envelope DAC raised-cosine ramp from 0 $\rightarrow$ CODE_FULL.

Key-up sequence: (1) envelope DAC ramps to 0 over 3–5 ms; (2) wait 1 ms guard after envelope nulls; (3) bias DACs commanded OPERATE $\rightarrow$ IDLE; (4) wait 200 μs to confirm tubes back in cutoff.

Fail-Safe

Three independent layers:

(1) DAC EEPROM safe-park: startup value 0 $\rightarrow$ IDLE bias on cold boot, before MCU initializes I²C.

(2) Watchdog gate: esp_task_wdt pulls a hardware GPIO that removes screen voltage if the MCU hangs. PA goes dark even if the DAC is stuck in OPERATE.

(3) Cathode-current monitor: independent ADC-based sense (separate circuit) triggers a soft-shutdown if either tube exceeds its current limit. Captured separately in the cathode-sense design notes.

Layout Notes

- Keep the op-amp/transistor output trace short and away from RF; the 22 k Ω grid leak provides isolation but route bias on a separate layer from RF signals where possible.
- -90 V rail is small-current but high-voltage. Use $\geq 150\text{ V}$ rated film/ceramic capacitors for bypass. Standard 25 V ceramics will fail.
- Allocate distinct I²C addresses for the two per-tube DACs (MCP4725 supports 0x60–0x67 via the A0/A1/A2 pads). Collisions kill the I²C bus.

• **Design B only:** mount the PNP transistor close to its base resistor and decouple its emitter with 100 nF to its supply pin side. The PNP is fast ($f_T \approx 50\text{ MHz}$) and can ring without proper damping if the base is driven through long inductive traces.

Design A vs. Design B Comparison

	Design A (OPA454)	Design B (Discrete PNP)
Op-amp	OPA454 (HV)	LM358 (LV) + MPSA92 PNP
Cost	~\$5 (single-source TI)	~\$0.50 (LM358 + MPSA92)
Supply margin	5 V under op max (100 V) — adequate	Unlimited (PNP $V_{CEO} = 300\text{ V}$)
Parts count	5 R + 1 IC + 1 ref + bypass	6 R + 1 IC + 1 transistor + 1 ref + bypass
Convenience	Single IC handles everything	Two-stage; transistor needs care for HF stability
Counterfeit risk	OPA454 has known counterfeit sources	MPSA92 / LM358 are generic — no incentive to fake
Authenticity check	Power test before installing; check the laser-etch	Normally, genuine vs. relabel can be hard to tell